Lesson 2 Action Group Programming

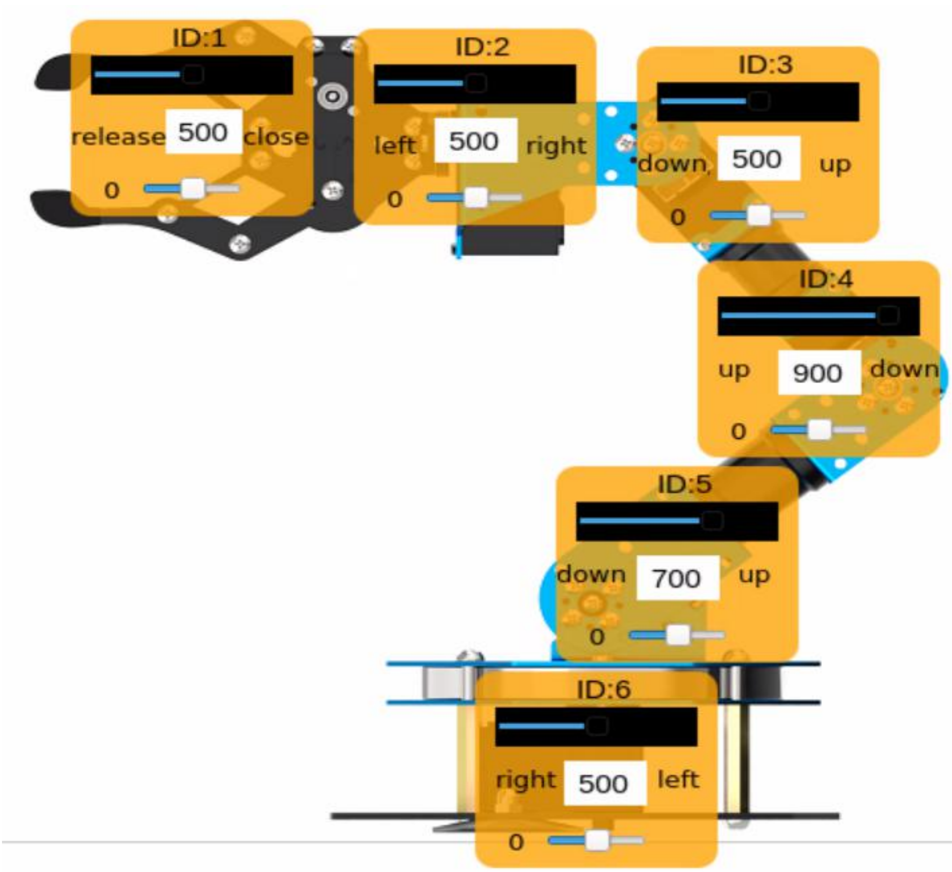
1. Project Outcome

Program an action group consisting of 12 actions to allow the ArmPi FPV to "downward to grip the item and place it in the front".

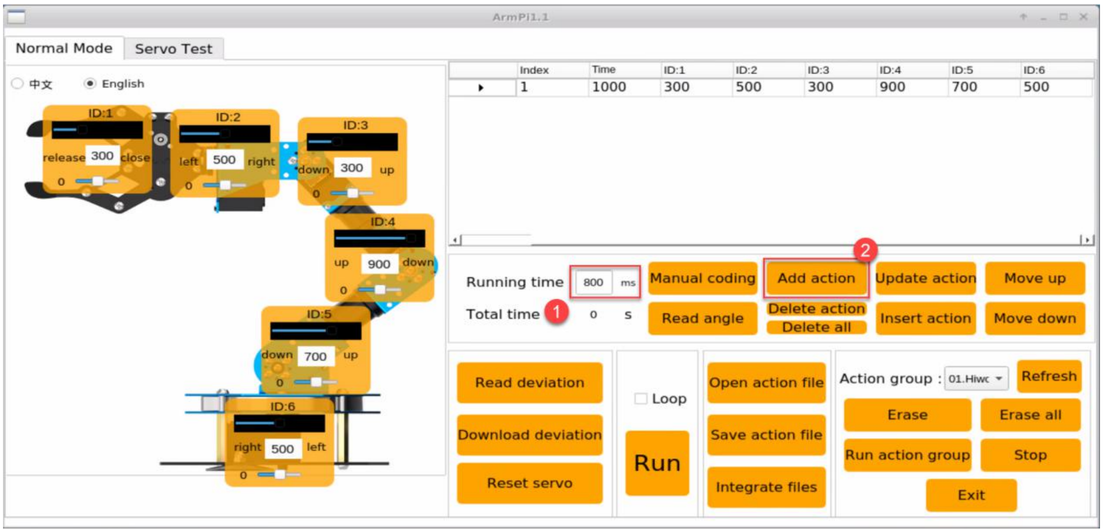
2. Complete Program

2.1 Action Programming

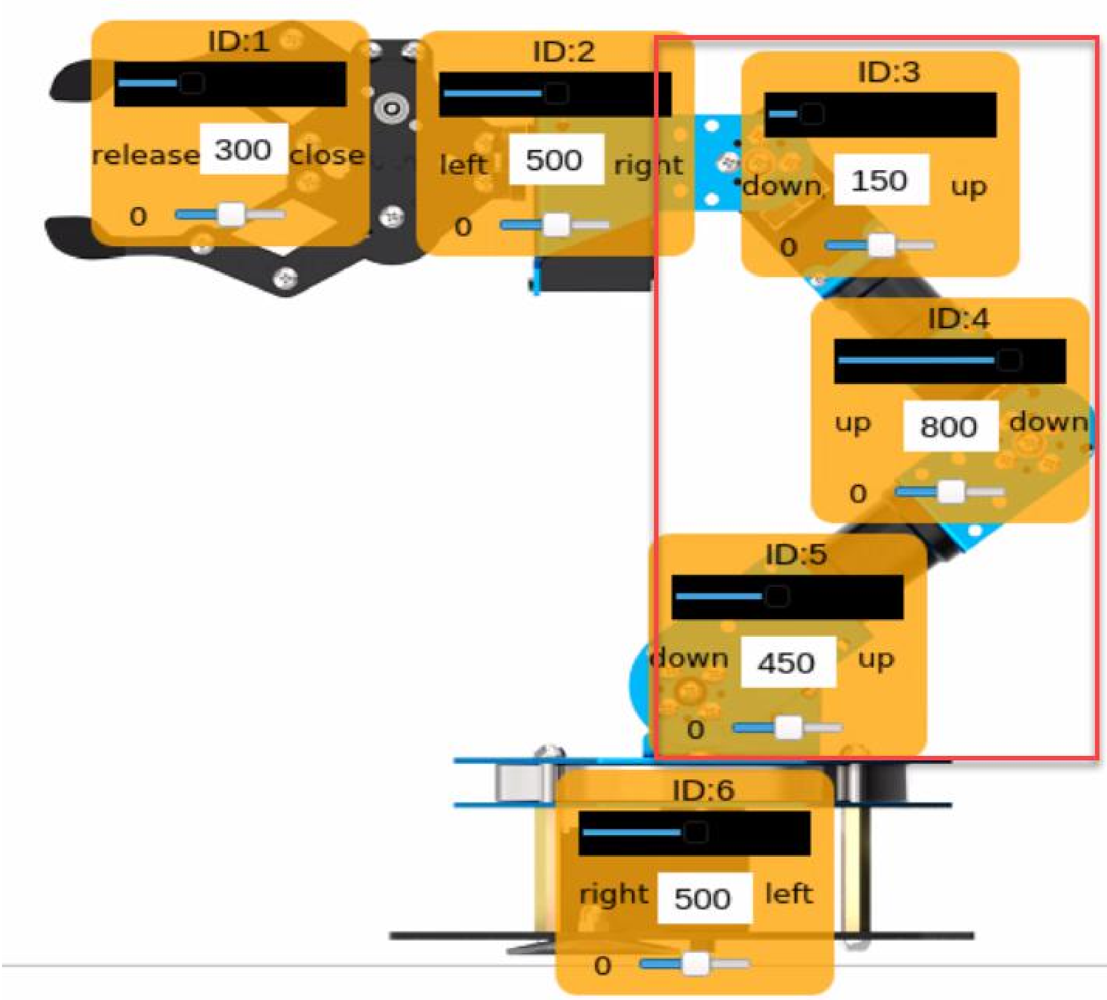
Step 1: first, we need to drag the slider to let the robotic arm return to original posture. Then click "Add action" to add this action to the action list.



Step 2: set the running time as 800ms and click “update action”



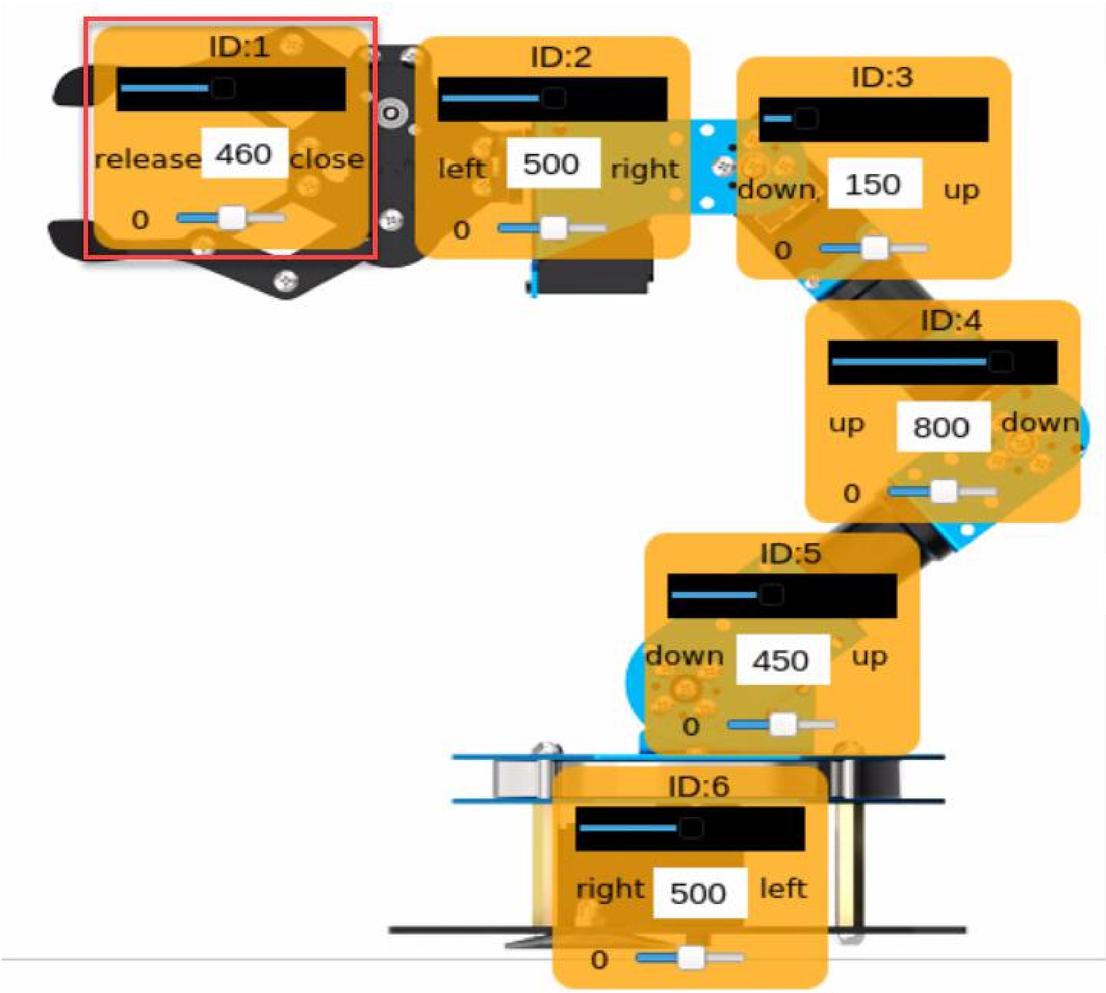
Step 3: Program to lower the arm. Drag the sliders of the No. 3, 4, 5 servos as the figure below. Then click “Add action” to add the action and the running time should be set as 800ms as well.



Step 4: add this action group to make the actions more coordinate and natural. Select the No.2 action and click "Add Action". The running time does not need to be set too long and you can modify it to 200 ms.

Index	Time	ID:1	ID:2	ID:3	ID:4	ID:5	ID:6
1	800	300	500	300	900	700	500
2	800	300	500	150	800	450	500
3	200	300	500	150	900	450	500

Step 5: Open the gripper. Drag the ID:1 slider to set the figure below, then click "add action" to add the fourth action. The running time should be set as 400ms.

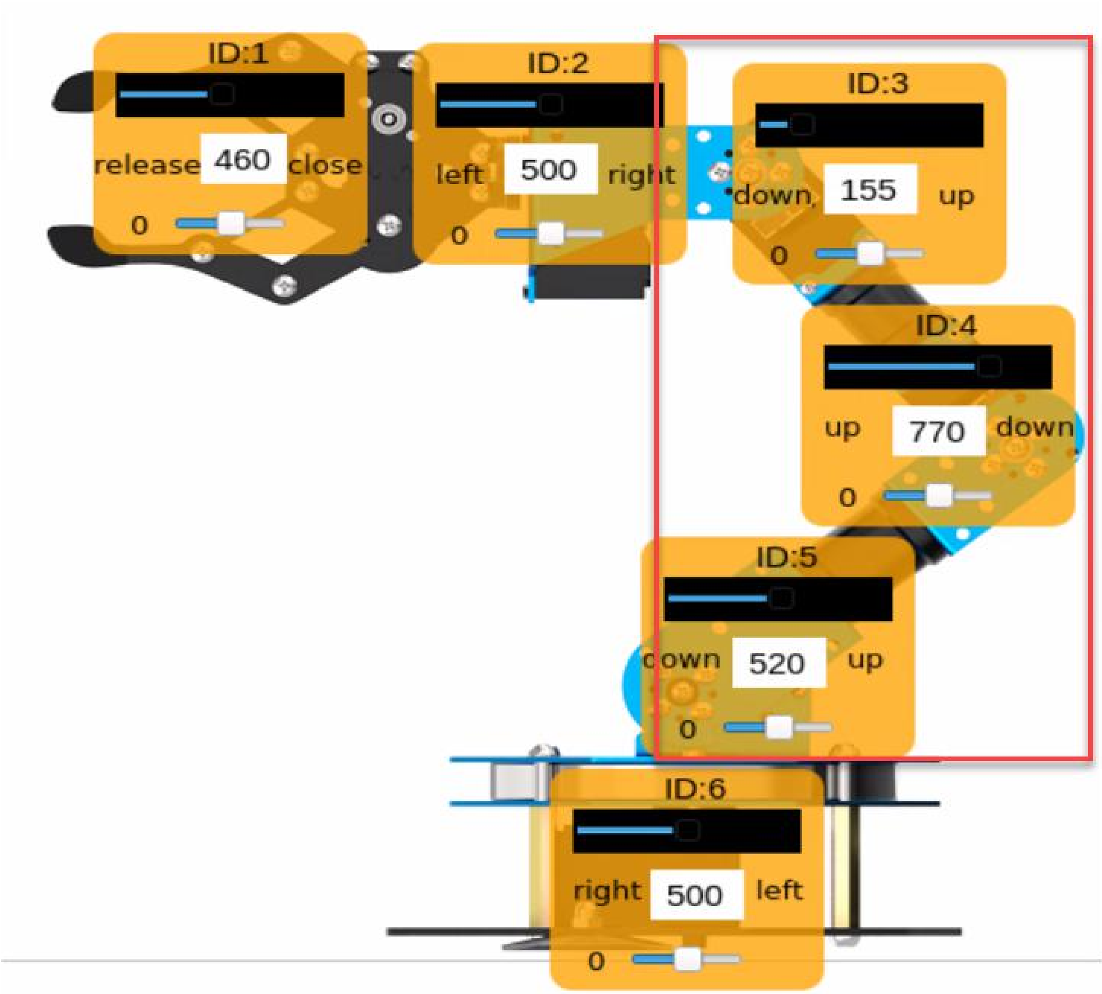


Step 6: In order to make the gripping action more stable, we make No.5 action transition action. Select the fourth action group, then click "add action" and set the running time as 200ms.

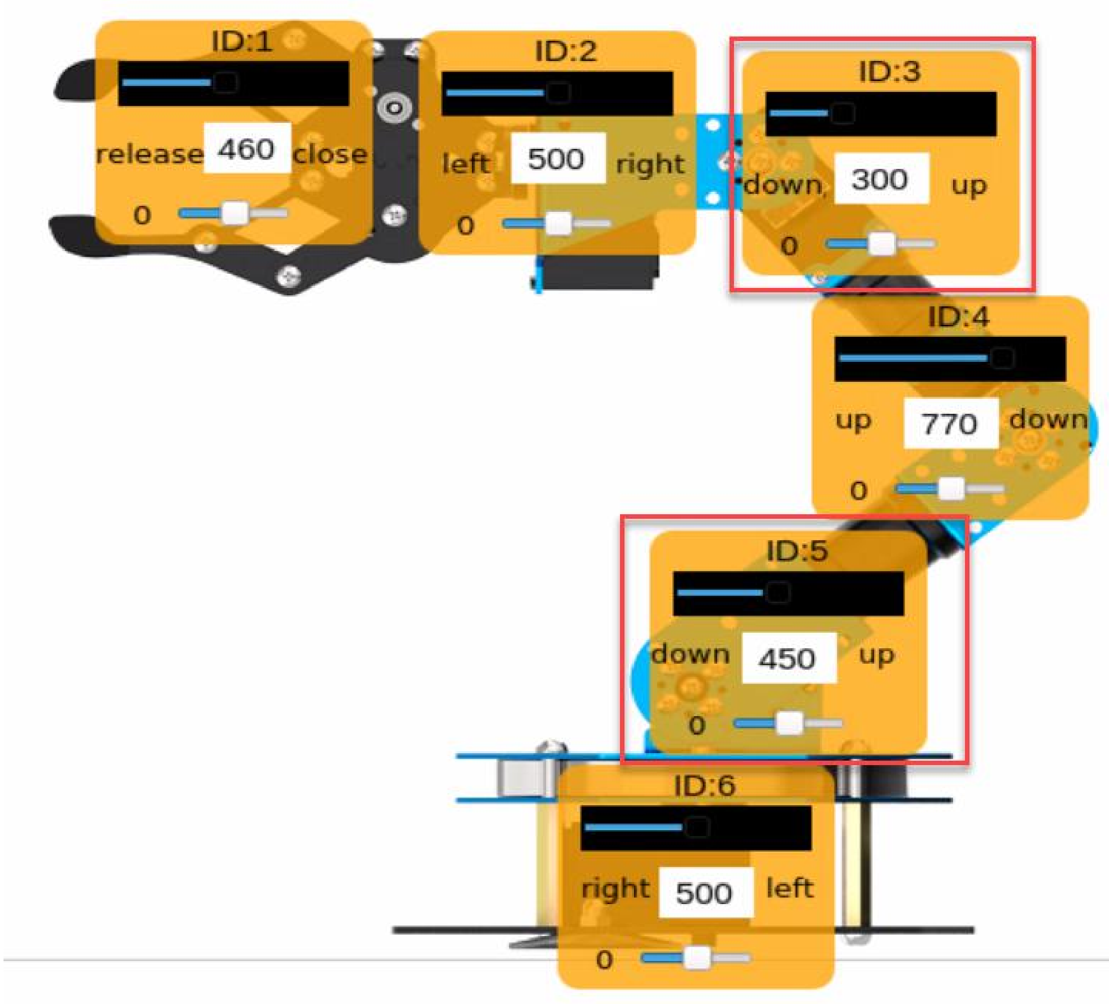
	Index	Time	ID:1	ID:2	ID:3	ID:4	ID:5	ID:6
	1	800	300	500	300	900	700	500
	2	800	300	500	150	800	450	500
	3	200	300	500	150	900	450	500
	4	200	460	500	150	800	450	500
▸	5	200	460	500	150	800	450	500

Step 7: Program the robotic arm to lift the item up after picking. Drag the

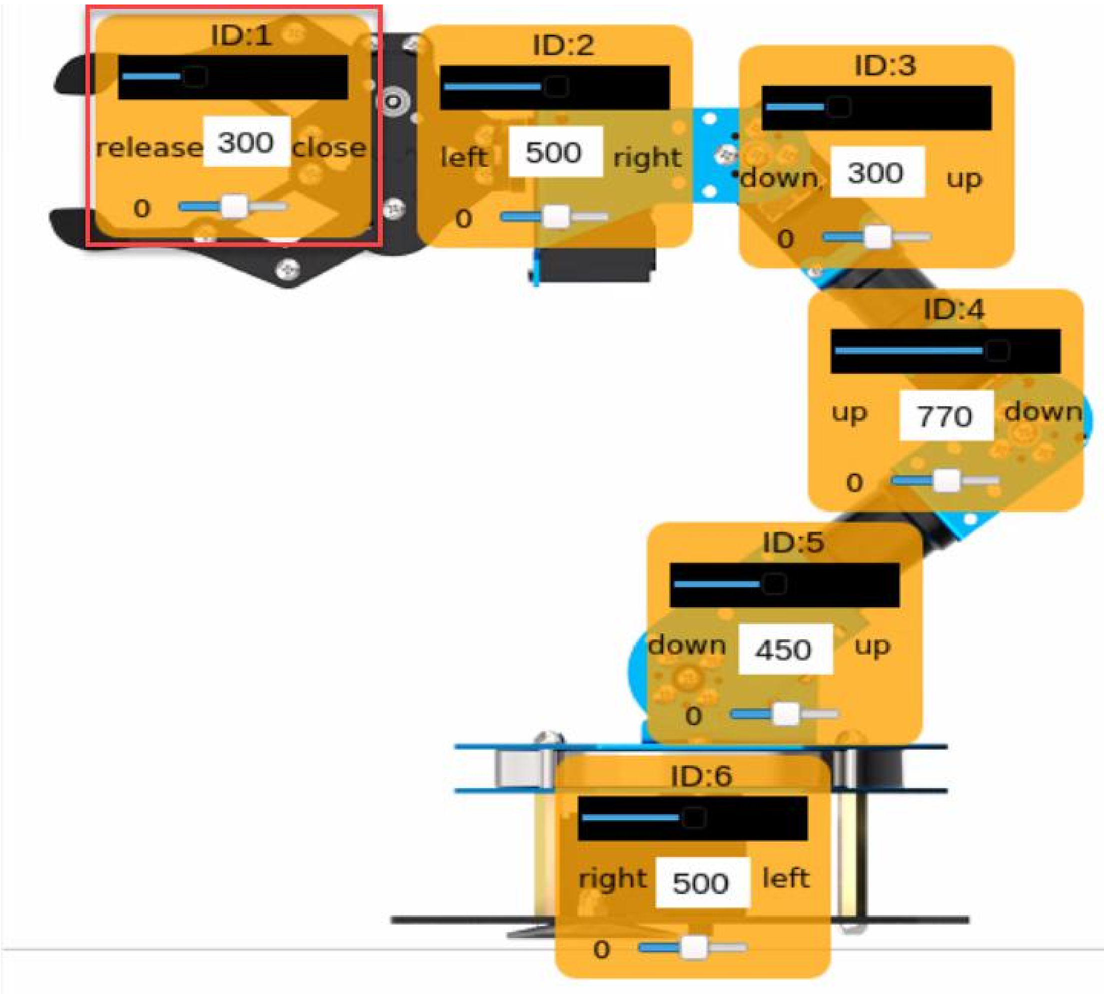
ID:3, ID:4, ID:5 slider in turn. Then set the running time as 500ms and add the action to the sixth action group.



Step 8: The 7th action group serves as a transition of the 6th action group, and the running time can be set to 200ms. Then we need to move the items forward. The adjusting figure of the 8th actions are as follow, and the running time is 800ms.



Step 9: when the robotic arm arrives at the set position, add the ninth action group as the transition group for the eighth action group, and the running time is set as 200ms. Then adjust ID:1 servo and set the running time as 600ms.



Step 10: Program the tenth and eleventh action groups to release the gripper to finish the task. And the the twelfth action group aims at making the robotic arm return to the original posture. And you can click ▶ on the left side of the first action group. Then set the running time as 800ms and click “Add Action”.

The intact action list is as follow.

Index	Time	ID1	ID2	ID3	ID4	ID5	ID6
1	800	300	500	300	900	700	500
2	800	300	500	150	800	450	500
3	200	300	500	150	800	450	500
4	400	460	500	150	800	450	500
5	200	460	500	150	800	450	500
6	500	460	500	155	770	520	500
7	200	460	500	155	770	520	500
8	800	460	500	300	770	450	500
9	200	460	500	300	770	450	500
10	600	300	500	300	770	450	500
11	200	300	500	300	770	450	500
12	800	300	500	300	900	700	500

2.2 Save Action Group

After programming, please save the action groups for future debugging. Click "Save action file", choose the path as "**/home/ubuntu/ArmPi/ActionGroups**", name it (such as "grab-forward") and then click "Save" button.